II. Write short answers:

1. What do you understand by a loop? What is its significance?
2. What are the different ways to generate a loop?
3. Write down the general format of :
   (a) for loop (b) do-while (c) while loop [ICSE 2005 & 2008]
4. What is the purpose of using ?
   (a) break statement (b) continue statement in a program [ICSE 2011]
5. Distinguish between:
   (a) step loop and nested loop (b) while and do-while loop [ICSE 2005]
6. Explain the term loop with an example.
7. State one difference and one similarity between while and do-while loop. [ICSE 2005 & 2010]
8. Name two jump statements and their use. [ICSE 2010]
9. What is meant by an infinite loop? Give an example. [ICSE 2008]

III. Perform the following tasks as directed:

1. Rewrite the following for loops by using while and do-while loops:

(a)
```
for(i=1;j=1;i<=10;i++;j++)
{
System.out.println(i*j);
}
```

(b)
```
int p= 20;
for(k=p;k>=0;k-=2)
{
s+=k;
}
System.out.println("Sum = "+s);
```

(c)
```
int a = 37,b,c=0;
for(b=1;b<=a;b++)
{
if(a%b==0)
c=c+1;
}
```

```
if(c==2)
System.out.println(a + " is a prime number");
else
System.out.println(a + " is not a prime number");
```

2. Rewrite the following using for loop:

(a)
```
int i=1,f=1;
do
{
f=f*i;
i++;
}
while(i<=10);
System.out.println(f);
```

(b)
```
a=1;
while(a<=5)
{b=1;
while(b<=a)
{
System.out.print(b);
b++;
}
System.out.println();
a++;
}
```

IV. Predict the output:

1. What will be the output of the following code? [ICSE 2010]

```
int m=2;
int n=15;
for(int i=1;i<5;i++)
m++; - - n;
System.out.println("m="+m);
System.out.println("n="+n);
```

2. Analyze the following program segment and determine how many times the loop will be executed. What will be the output of the program segment?

```
int k=1,i=2;
while(++i<6)
k*=i;
System.out.println(k);
```
[ICSE 2010]

3. What will the following function return when executed?

(i) Math.max(-17,-19) (ii) Math.ceil(7.8) [ICSE 2009]

4. The following is a segment of a program.

```
x = 1; y = 1;
if(n>0)
{
x = x + 1;
y = y +1;
}
```

What will be the value of x and y, if n assumes a value:

(i) 1 (ii) 0 ? [ICSE 2009]

5. Analyze the following program segment and determine how many times the body of the loop will be executed (show the working).

```
x = 5; y = 50;
while (x<=y)
{
y = y/x;
System.out.println(y);
}
```

[ICSE 2009]

V. Java Programming:

1. Write a program to display the first ~~ten~~ n terms of the series:
   (a) 1, -3, 5, -7, 9, ...
   (b) 2, 5, 10, 17, ...
   (c) 0, 1, 2, 3, 6, ...
   (d) 0, 3, 8, 15, ...
   (e) 1, 12, 123, 1234, ...

   [ICSE 2003]
2. Write a program to find the factorial of 10.
   [Factorial of 10! = 10*9*8*7.....1]
3. Write a program to enter two numbers and check whether they are co-prime or not.
   Sample Input: 14, 15
   Sample Output: They are co-prime.
4. Write a program to find the factors of a number including 1 and the number itself.
   Sample Input: 18
   Sample Output: 1, 2, 3, 6, 9, 18
5. Write a program to accept a number and display the sum of its digits.
   Sample Input: 542
   Sample Output: 5 + 4 + 2 = 11
6. Write a program to print the sum of negative numbers, sum of positive odd numbers and sum of positive even numbers from a list of numbers entered by the user. The list terminates when the user enters zero. [ICSE 2005]
7. Write a program to calculate and print the sum of odd numbers and the sum of even numbers for the first n natural numbers. [ICSE 2006]
8. Write a program to compute and display the sum of the following series:
   $$S = \frac{1+2}{1*2} + \frac{1+2+3}{1*2*3} + \ldots\ldots\ldots\ldots + \frac{1+2+3+\ldots\ldots\ldots\ldots+n}{1*2*3\ldots\ldots\ldots\ldots*n}$$ [ICSE 2007]
9. Write a program to accept a number and display the new number after removing all zeros.
   Sample Input: 5400207
   Sample Output: 5427
10. Write a program to accept a number and display the frequency of each digit present in the number.
    Sample Input: 341124
    Sample Output: The frequency of 1 = 2
    The frequency of 4 = 2
11. Write a program to accept a number and check whether the number is present in the Fibonacci series or not. The program displays the message accordingly.
    Sample Input: 55
    Sample output: 55 is present in the Fibonacci series.

12. A prime number is said to be 'Twisted Prime', if the new number obtained after reversing the digits is also a prime number. Write a program to accept a number and check whether the number is 'Twisted Prime' or not.

    Sample Input: 167

    Sample Output: 761

    167 is a 'Twisted Prime'.

13. A number is said to be Unique if digits of a number are not repeated in it. Write a program to accept a number and check whether the number is Unique or not. The program displays the message accordingly.

    Sample Input: 5463

    Sample Output: It is a unique number.

    Sample Input: 3272

    Sample Output: It is not a unique number.

14. Write a program to accept two numbers and find the Lowest Common Multiple (LCM) of the numbers.

    [**Hint:** LCM = $\frac{\text{Product of two numbers}}{\text{HCF}}$]

15. Write a program to accept two numbers and check whether they are twin prime or not.

    [**Hint:** Twin prime numbers are the prime numbers whose difference is two.

    e.g.: (5, 7), (11, 13) .....]

16. A number is said to be Duck if the digit zero is (0) present in it. Write a program to accept a number and check whether the number is Duck or not. The program displays the message accordingly. (The number must not begin with zero)

    Sample Input: 5063

    Sample Output: It is a Duck number.

    Sample Input: 7453

    Sample Output: It is not a Duck number.

17. Write a program in Java to find the sum of the given series:

    (a) S = 9 + 99 + 8 + 89 + 7 + ... to n

    (b) S = 1 + 1 + 2 + 3 + 5 + ... to n terms

    (c) S = 2 – 4 + 6 – 8 + ... to n [ICSE 2008]

    (d) S = (1*2) + (2*3) + ... + (19*20)

    (e) S = 1 + (1+2) + (1+2+3) + ... + (1+2+3+ ..... + n) [ICSE 2003]

    (f) S = 1 + (1*2) + 2 + (1*2*3) + 3 + ... + 9 + (1*2*3* ..... 10)

    (g) S = 1! + 2! + 3! + 4! + ... to n terms

    (h) S = $\frac{1}{2} + \frac{2}{3} + \frac{3}{4}$ + ... to n terms

    (i) S = $1 + \frac{3}{4} + \frac{5}{9}$ + ... to n terms

    (j) S = $\frac{2}{3} - \frac{4}{5} + \frac{6}{7} - \frac{8}{9}$ + ... to n

18. Write a program to find the sum of the given series:

    (a) S = $a + \frac{a^2}{2} + \frac{a^3}{3}$ + ... + $\frac{a^n}{n}$

    (b) S = $a + \frac{a}{2!} + \frac{a}{3!}$ + ... + $\frac{a}{n!}$

(c) $S = \frac{a}{2} + \frac{a}{4} + \frac{a}{8} + \ldots\ldots\ldots\ldots$ to n

(d) $S = 1 - \frac{a}{2} + 3 - \frac{a}{4} + 5 - \frac{a}{6} + \ldots\ldots\ldots\ldots$ to n

(e) $S = \frac{a}{2!} - \frac{a}{3!} + \frac{a}{4!} - \frac{a}{5!} + \ldots\ldots\ldots\ldots + \frac{a}{20}$

(f) $S = (a+1) + (a+2) + (a+3) + \ldots\ldots\ldots\ldots + (a+n)$

(g) $S = \frac{a+1}{3} + \frac{a+2}{5} + \frac{a+3}{7} + \ldots\ldots\ldots\ldots$ to n

(h) $S = \frac{a}{2} + \frac{a}{5} + \frac{a}{8} + \frac{a}{11} + \ldots\ldots\ldots\ldots + \frac{a}{20}$ [ICSE 2008]

(i) $S = \frac{1}{a} + \frac{2}{a^2} + \frac{3}{a^3} + \ldots\ldots\ldots\ldots + \frac{n}{a^n}$

(j) $S = a - \frac{a^3}{5} + \frac{a^5}{9} - \frac{a^7}{13} \ldots\ldots\ldots\ldots$ to n

19. Write a program to display the following pattern:

(a)
```
1
3 1
5 3 1
7 5 3 1
9 7 5 3 1
```

(b)
```
1 2 3 4 5
6 7 8 9
10 11 12
13 14
15
```

(c)
```
15 14 13 12 11
10 9 8 7
6 5 4
3 2
1
```

(d)
```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

(e)
```
1 2 3 4 5
2 2 3 4 5
3 3 3 4 5
4 4 4 4 5
5 5 5 5 5
```

(f)
```
1 3 5 7 9
3 5 7 9 1
5 7 9 1 3
7 9 1 3 5
9 1 3 5 7
```

20. On the basis of sale, a pharmaceutical company announces the following new tariff of commission for their Medical Representatives and Distributors.

Sale	Commission of Representative	Commission of Distributor
Up to ₹ 10,000	5%	2%
₹ 10,001–₹ 20,000	8%	3%
₹ 20,001–₹ 30,000	10%	4%
More than ₹ 30,000	12%	5%

Write a program to accept monthly sale from 50 different representatives along with their distributors. Calculate their commissions and display the output in the given format:

Sl. No.	Sale	Commission of Representative	Commission of Distributor
xxx	xxx	xxx	xxx

21. In an election, out of 1249 voters in a booth, only 861 voters used their franchise correctly. If five candidates are contesting, write a program in Java to find:

(a) the number of invalid and valid votes and

(b) the percentage of the valid votes received by each candidate. [ICSE 1999]

22. Computech company has 96 employees who are divided into 4 grades as per their basic pay as the following:

GRADE I BASIC	₹ 10,000 P.M. or more
D.A.	40% of Basic
H.R.	30% of Basic
GRADE II BASIC	₹ 5,000 P.M. or more but less than ₹ 10,000
D.A.	40% of Basic
H.R.	25% of Basic
GRADE III BASIC	less than ₹ 5,000 but more than ₹ 2,000 P.M.
D.A.	30% of Basic
H.R.	20% of Basic
GRADE IV BASIC	₹ 2,000 P.M. or less
D.A.	30% of Basic
H.R	15% of Basic

If the salary, which is the total of Basic, D.A. and H.R. is above ₹ 50,000 per annum then income tax at the rate of 30% of the annual salary exceeding ₹ 50,000 is deducted on monthly basis at source. Taking name of the employee and the basic (monthly) pay as inputs a pay slip for each employee is to be printed. Write a program in Java to perform the job. [ICSE 1995]

23. A computerized ticket counter of an underground metro rail station charges for each ride at the following rate:

Age	Amount/Head
18 or above	₹ 5
5 or above but below 18	₹ 3
Accompanying kids below 5	NIL

Write a program in Java, which takes as input the number of people of various age groups and prints a ticket. At the end of the journey, the program states the number of passengers of different age groups who travelled and the total amount received as collection of fares. [ICSE 1995]

24. The sum of the cubes of the natural numbers is given by the following formula:

$$1^3 + 2^3 + 3^3 + \ldots\ldots\ldots\ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

It is intended to write a program to compute the value of S, where

$$S = 3^3 + 4^3 + 5^3 + \ldots\ldots\ldots\ldots + n^3$$

(a) State: (i) the variable whose value is to be taken as input.
(ii) the variable whose value is to be obtained as output.

(b) Break up the processing job into four sequential steps and write those steps.

(c) Write a program in JAVA to find S. [ICSE 2000]

25. A number of answer scripts with scores 0 to 99 awarded after marking are to be searched for finding:
    (a) Number of answer scripts marked;
    (b) Percentage of candidates getting 85 and above;
    (c) Percentage of candidates getting 39 or less.
    (d) Write a Java program to take the scores as inputs and then output the three quantities required. [ICSE 1998]
26. A game of throwing dice is played between two players in which each player throws a dice unless his score adds up to 20. A player is declared 'Winner' with the minimum number of throws. Write a program to perform the task given above.
27. Write a program to simulate a game of throwing dice 10 times, 100 times and 1000 times. Print the score obtained in each category.
28. A class teacher wants to know the performance of the students of her class in Science (Physics, Chemistry, Biology). Write a program to accept marks in Physics, Chemistry and Biology of each student. Calculate the average and count how many of them have secured 90% or more (in average) in the class having 45 students.
29. Write a program to generate a triangle or an inverted triangle till n terms based upon User's choice of triangle to be displayed. [ICSE 2009]

Example 1:

Input: Type 1 for a triangle and
Type 2 for an inverted triangle
Enter your choice 1
Enter the number of terms 5

Example 2:

Input: Type 1 for a triangle and
Type 2 for an inverted triangle
Enter your choice 2
Enter the number of terms 6

Sample Output:

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
6 6 6 6 6 6
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

30. Write a menu driven program to accept a number from the User and check whether it is a 'BUZZ' number or to accept any two numbers and print the 'GCD' of them.
    1. A BUZZ number is the number which either ends with 7 or is divisible by 7.
    2. GCD (Greatest Common Divisor) of two integers is calculated by continued division method. Divide the larger number by the smaller, the remainder then divides the previous divisor. The process is repeated till the remainder is zero. The divisor then results the GCD. [ICSE 2009]
31. Write a menu driven program to accept a number from the user and check whether it is a Prime number or an Automorphic number.
    (i) Prime number: (A number is said to be prime, if it is divisible by 1 and itself)
    Example: 3,5,7,11..... .
    (ii) Automorphic number: (Automorphic number is the number, which is contained in the last digit(s) of its square.)
    Example: 25 is an Automorphic number as its square is 625 and 25 is present as the last two digits. [ICSE 2010]

32. Write a menu driven program to perform the following tasks by using Switch case statement:

1. To print the series:

   0,3,8,15,24, ... to n terms. (value of 'n' is to be an input by the user)

2. To find the sum of the series:

   $S = \frac{1}{2} + \frac{3}{4} + \frac{5}{6} + \frac{7}{8} + \ldots\ldots\ldots\ldots + \frac{19}{20}$ [ICSE 2011]

33. Write a program to input a number and print whether the number is a special or not. A number is said to be special number, if the sum of the factorial of the digits of the number is same as the original number.

Sample Input: 145 is a special number because 1! + 4! + 5! = 1 + 24 + 120 = 145

where, ! stands for factorial of the number and the factorial value of a number is the product of all integers from 1 to that number.

e.g.: 5! = 1*2*3*4*5 = 120 [ICSE 2011]

34. Using a switch statement write a menu driven program to:

(i) Generate and display the first 10 terms of the Fibonacci series

0, 1, 1, 2, 3, 5 .....

The first two Fibonacci numbers are 0 and 1, and each subsequent number is the sum of the previous two.

(ii) Find the sum of the digits of an integer that is input.

Sample Input: 15390

Sample Output: Sum of the digits = 18

For an incorrect choice, an appropriate error message should be displayed.

[ICSE 2012]